

Project Initialization and Planning Phase

Date	26 June 2026
Team ID	SWIID-2026-1243
Project Title	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to enhance the learning experience by developing **EduGenie**, an AI-powered educational assistant that provides intelligent question answering, simplified concept explanations, AI-generated quizzes, educational text summarization, and personalized learning recommendations. By integrating **Google Gemini 1.5 Pro**, **LaMini-Flan-T5**, and **FastAPI**, the system delivers accurate, interactive, and personalized educational support through a single platform.

Project Overview	
Objective	The primary objective of EduGenie is to simplify learning by using Generative AI to provide accurate educational assistance, improve understanding of concepts, generate quizzes, summarize study materials, and recommend personalized learning paths for students and self-learners.
Scope	The project focuses on developing a web-based AI educational assistant that supports multiple learning features, including intelligent question answering, concept explanation, quiz generation, text summarization, and personalized learning recommendations through an interactive and user-friendly interface.

Problem Statement

Description Students and self-learners often spend significant time searching across multiple websites and educational platforms to understand concepts, prepare quizzes, summarize notes, and plan their learning. The lack of an integrated educational platform makes learning inefficient and time-consuming.

Impact By solving these challenges, EduGenie will improve learning efficiency, reduce the time required to access quality educational resources, increase

student engagement, and provide a personalized learning experience that supports better academic performance.

Proposed Solution	
Approach	Develop an AI-powered educational assistant using FastAPI , Google Gemini 1.5 Pro , and LaMini-Flan-T5 . The system will process user queries using prompt engineering and NLP techniques to generate accurate educational responses, quizzes, summaries, and personalized learning roadmaps.
Key Features	● ntelligent Question Answering ● Simplified Concept Explanation ● AI-Powered Quiz Generation ● Educational Text Summarization ● Personalized Learning Path Recommendations ● Interactive and Responsive User Interface ● FastAPI-based REST APIs ● 🎬 Integration with Google Gemini 1.5 Pro and LaMini-Flan-T5



Resource Requirements

Resource Type	Description	Specification/Allocation
	Computing Resources	or higher (Mac M1 also Intel i5/Ryzen 5 supported) Memory Minimum 8 GB RAM
		Storage 256 GB SSD or higher
Hardware		CPU/GPU specifications, number of cores T4 GPU
Computing Resources		

Memory RAM specifications 8 GB Storage Disk space for data, models,

and logs 1 TB SSD

Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	Development Environment	Visual Studio Code / PyCharm
Data		
Data	Educational Data	User questions, educational text, study materials, AI-generated responses